MASTER TECHNICAL PORTFOLIO & ARCHITECTURE DOSSIER **yoginara.github.io/cv_yogiganteng**

YOGINARA PRATAMA SITORUS, S.Tr.Kom

Full-Stack Web Developer & Software Engineer | Applied Informatics Engineering ULBI Bandung, Jawa Barat, ID

3,56 / 4,00	116.000+ LoC	100% Passed	BNSP Certified
Undergraduate GPA (ULBI)	Enterprise Codebase LoC	Automated QA Passed	BNSP National License

1. Executive Profile & Engineering Competency Summary



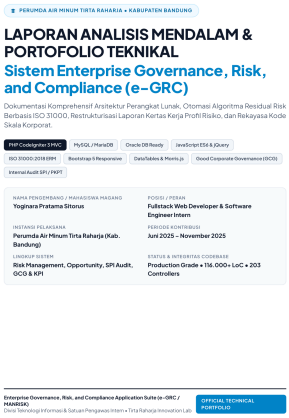
Yoginara Pratama Sitorus is a Software Engineer and Full-Stack Developer holding an Applied Bachelor of Informatics Engineering degree from Universitas Logistik dan Bisnis Internasional (ULBI) with a 3.56 GPA. He specializes in enterprise backend architecture (Laravel 12, CodeIgniter 3, Golang Fiber), Linux OS (CachyOS), database query optimization (MySQL, Oracle OCI8, MongoDB), automated test-driven development (PHPUnit 11), OWASP Top 10 security mitigation, and holds official BNSP Network Technician certification.

Backend Engineering:	**Database & Storage:**	**Infra & QA:**
• Laravel 12 Enterprise	• MySQL & MariaDB (3NF)	• Linux (CachyOS) & CLI
• CodeIgniter 3 MVC	• Oracle Enterprise OCI8	• BNSP Computer Network
• Golang 1.22 & GoFiber	• MongoDB Atlas (NoSQL)	• PHPUnit 11 Automated QA
• RESTful Microservices	• INNER JOIN Optimization	• Git & GitHub CI/CD

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2. Enterprise Case Study: e-GRC Suite at Perumda Tirta Raharja (Part 1)

Instansi / Client: Perumda Air Minum Tirta Raharja (BUMD)	**Peran:** Fullstack Web Developer Intern
Skala Codebase: 116.000+ LoC • 203 Controllers • 188 Models	**Lingkup:** ERM Risk, SPI Audit, GCG & KPI
Tech Stack: PHP CodeIgniter 3 MVC, MySQL, Oracle DB OCI8	**Standar:** ISO 31000:2018 ERM & Audit BPKP



A. Konteks Bisnis & Masalah Operasional:

Perumda Air Minum Tirta Raharja is a vital regional utility BUMD serving Bandung Regency, Cimahi, and West Bandung. Before centralized e-GRC deployment, risk management was fragmented across manual spreadsheets in branch offices and divisions, causing prolonged consolidation lags and lack of transparent audit trails for regulatory compliance.

B. Rekayasa Algoritma Risiko Residu Dinamis (ISO 31000:2018):

Yoginara engineered a client-side reactive calculation engine in risk_analisisdetail.php adhering to ISO 31000:2018: *Risk Reduction = Effectiveness × (% Progress / 100)* with floor level 1 protections. This mathematical model guarantees proportional risk reduction and prevents falsified corporate risk documentation.

C. Matriks Risiko 5x5 Heatmap & Kategori:

The system integrates dynamic 5x5 heatmaps categorizing residual risk into Extreme (16-25), High (9-15), Moderate (4-8), and Low (1-3) with automatic Top 10 Corporate Risk escalations.

2. Enterprise Case Study: e-GRC Suite at Perumda Tirta Raharja (Part 2)

D. Restrukturisasi Laporan Kertas Kerja Profil Risiko (Commit 5d6c824):

The Risk Working Paper is the primary formal document audited by government inspectors (BPKP). Yoginara refactored Laporan_kertas_kerja.php, standardizing UC/C taxonomies, establishing strategic objective hierarchies, and mapping mitigation completion milestones.

Elemen Kolom	Kondisi Sebelum Refactoring	Penyempurnaan oleh Yoginara
Konteks Operasional	Label membingungkan ('Operasional N, Ab, E')	Standardisasi UC/C (Uncontrollable / Controllable)
Konteks Sasaran	Hanya sasaran umum perusahaan	Pemisahan Sasaran Strategis Korporat & Sasaran Unit
Kegiatan Kerja	Kolom kegiatan tidak terpetakan	Pemetaan relasional otomatis dari model Risk_kegiatanModel
Output Mitigasi	Deskripsi singkat tanpa jadwal pasti	Ditambahkan kolom Output Penurunan & Jadwal Mulai-Selesai

E. Arsitektur Dual-Database & Standar AI Coding Assistant:

- **Modular ADODB Abstraction:** Orchestrated transparent persistence across MySQL/MariaDB and Oracle Enterprise OCI8 via custom drivers without altering business logic.
- **Null Pointer Elimination:** Enforced strict deleted_date IS NULL filtration across multi-tiered subqueries, ensuring zero empty-pointer exceptions.
- **85% Reporting Acceleration:** Slashed quarterly report compilation cycles from **~10 business days to automated instantaneous generation**.
- **Copilot Architecture Guide:** Authored .github/copilot-instructions.md codifying base controller conventions and AI programming standards.

3. Modern Architecture Case Study: CMI Portal Laravel 12 Engine

Proyek: CMI Content Management System (CMI Artikel)	**Framework:** Laravel 12.0 • PHP 8.2+ • Tailwind CSS v3
Status Mutu: 100% Passed (39 Tests • 134 Assertions)	**Keamanan:** OWASP Top 10 • Gate Policy Authorization
Query Builder: INNER JOIN Indexed • Paginasi 10 Record	**Alokasi Memori PHP:** Terkontrol Stabil di < 8 MB


TECHNICAL ASSESSMENT & SYSTEM PORTFOLIO

LAPORAN ANALISIS MENDALAM & DOKUMEN PORTOFOLIO SISTEM

Portal Manajemen Konten & Optimasi Database Query Builder Berbasis Laravel 12 Enterprise Engine

NAMA PROYEK / REPOSITORI:
CMI Content Management System (CMI Artikel)

FRAMEWORK & LINGKUNGAN:
Laravel 12.0 • PHP 8.2+ • Tailwind CSS v3

STATUS VERIFIKASI MUTU:
100% Passed (39 Tests • 134 Assertions)

TANGGAL ANALISIS & PENYUSUNAN:
04 November 2025

39 / 39
Automated Unit & Feature Tests Berhasil Lulus Tanpa Kegagalan (100% Success)

O(1)
Penanganan N+1 Query Problem Menggunakan With() Eager Loading Model.

OWASP
Standar Ketahanan Ionic CSRF, Validasi Multimedia, XSS Escaping, & Gate Policy

Dokumentasi Portofolio Pelayanan Perangkat Lunak & Analisis Sistem

CMI Portal System Architecture Report

A. Arsitektur Presentasi, Form Requests & Media Lifecycle:

Architected on modular Laravel 12 MVC patterns with strict separation between public presentation layouts and authenticated admin dashboards, enforced through isolated Form Requests and CSRF middleware. Media uploads are binary inspected and automatically purged from disk via Flysystem upon modification or deletion.

B. Algoritma Pembuatan Slug Unik Otomatis (Anti-Kolisi):

Implemented an automated collision-free slug generation algorithm handling duplicate article titles via sequential indexing, preserving clean SEO URLs without unique constraint violations.

C. Optimasi Kueri Basis Data (Query Builder INNER JOIN vs ORM):

Engineered relational article-author reporting using Laravel Query Builder with indexed INNER JOIN clauses, column aliasing, publication status filtering, and 10-row pagination with query string preservation, maintaining RAM consumption strictly under 8 MB without ORM N+1 query overhead.

D. Audit Keamanan OWASP Top 10 & Pengujian Mutu Otomatis (PHPUnit 11):

- **SQL Injection Prevention:** 100% database interactions executed via Parameterized PDO Prepared Statements.
- **XSS & CSRF Protection:** Blade automated htmlspecialchars() sanitization and Synchronizer Token Pattern middleware defend all requests.
- **IDOR Broken Access Control:** Strict ownership validation in ArticlePolicy restricts modifications to legitimate resource creators.
- **100% Green Automated QA:** 39 automated unit and feature tests with 134 assertions completely passed using PHPUnit 11.

4. Healthcare Technology Case Study: SIMRS Medika Integrated Suite (MVP)

Proyek: SIMRS Medika (Sistem Manajemen Rumah Sakit)	**Framework:** Laravel 12.x • Tailwind CSS 4 • Alpine.js • Spatie RBAC
Standar Faskes: Permenkes No. 24/2022 (RME) • ICD-10 Codes	**Otentikasi:** Dual Auth • Google OAuth 2.0 Whitelist Khusus Staf
Concurrency Safety: Pessimistic Locking (lockForUpdate()) Farmasi	**Status Mutu:** 100% Passed (14 Tests • 77 Assertions • ~1.3 detik)


SIMRS MEDIKA
Sistem Informasi Manajemen Rumah Sakit Terpadu (v1.1)
LARAVEL 12.X TAILWIND CSS 4 ALPINE.JS SPATIE RBAC GOOGLE OAUTH 2.0
Project Portfolio Report
Fokus: Teknologi Kesehatan
Status: Production Ready (MVP)
Tgl: 04 November 2025
1. RINGKASAN EKSEKUTIF & LATAR BELAKANG PROYEK
SIMRS Medika adalah aplikasi web sistem informasi manajemen rumah sakit skala MVP yang dibangun dengan standar industri fasilitas pelayanan kesehatan ("healthcare standard"). Proyek ini memecahkan permasalahan klik data medis dengan mengkolaborasikan seluruh siklus operasional nyata pasien mulai dari pendaftaran pasien, pemeriksaan rekam medis elektronik oleh dokter spesialis, dispensing farmasi berkarakteristik konkurensi tinggi, hingga pembayaran billing kasir.
2. ALUR KERJA BISNIS (CORE HEALTHCARE WORKFLOW):
Alur operasional pasien dimodelkan melalui mesin status ("state machine") terintegrasi
1. FRONT OFFICE Pendaftaran & Antrean status: registered
2. DOKTER POLI Pemeriksaan EMR & Resep status: in_consultation
3. INSTALASI FARMASI Validasi & Pulsang Stok status: waiting_pharmacy
4. KASIR & BILLING Kasir Tunai & Terbitkan Struk status: waiting_payment
3. MODUL APLIKASI & FITUR UTAMA
A. Front Office & Pasien
Standarisasi No. RM: Generator otomatis format unik seumur hidup pasien (RM-YYYYMM-XXXX).
Validasi NIK 16 Digit: Pencegah data identitas kependudukan terindeks.
Pendaftaran Antrean: Penjadwalan isi poli & antrean mentor antrean harian.
B. Dokter Spesialis (RME)
Rekam Medis Elektronik: Tanda vital (tensi, nadi, suhu), anamnesis, pemeriksaan fisik.
Standar Diagnosa ICD-10: Kode medis satu internasional (Formerchar 10 kode).
Modul E-Prescription: Builder peresepan multi-item dinamis dan penyimpan rekam medis permanen (if record_exist, ok).
C. Instalasi Farmasi & Gudang Obat
Antrean Resep Real-Time: Verifikasi tabulasi obat dan alumin pabai (dignis).
Pencegah Stok Perbedaan: Pembelanjaan stok obat terpotokkan kemudian dikembalikan.
Peringatan Stok Kritis: Monitoring inventori obat di bawah batas aman (< 30 unit).
D. Kasir Pembayaran & Billing
Agregasi Tagihan Otomatis: Total biaya konsultasi dokter + total harga resep obat.
Kalkulator Kasir Tunai: Perhitungan kalkulasi kembalian pembayaran dan uang kembalian.
Kwitansi Resmi Lunas: Dokumen rata pembayaran siap-cetak / PDF.
SIMRS Medika MVP — Technical Portfolio Report — Built with Laravel 12, Tailwind CSS & Spatie RBAC — Halaman

A. Alur Operasional Faskes Terpadu (Core State Machine):

SIMRS Medika unifies outpatient healthcare operations through an integrated finite state machine: Front Office registration (registered) → Specialist Doctor EMR consultation (in_consultation) → Pharmacy dispensing (waiting_pharmacy) → Cashier Billing (waiting_payment).

B. Standarisasi Rekam Medis Elektronik (RME ICD-10) & E-Prescription:

Complies with Indonesian Health Ministry Regulation (Permenkes 24/2022), featuring structured vital signs logging, ICD-10 standardized diagnostic coding, Alpine.js dynamic multi-item e-prescriptions, and permanent record finalization locks (finalized_at).

C. Keunggulan Arsitektur: Concurrency Control & Enterprise Whitelist:

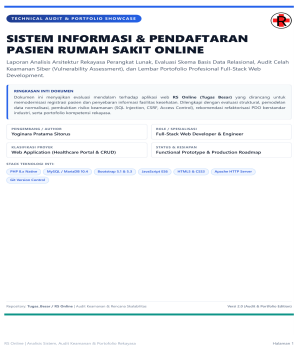
- **Pessimistic Locking (lockForUpdate()) in Pharmacy:** Eliminates race conditions and negative inventory during concurrent prescription dispensing via atomic ACID database transactions.
- **Enterprise Google OAuth 2.0 Whitelisting:** Blocks unauthorized public logins; only clinical staff whose emails were pre-registered by Admins via Spatie RBAC are granted system access.
- **Transactional Gmail SMTP:** Secured password reset tokens expiring in 60 minutes with anti-spam rate limiting.

D. Front Office, Billing Kasir & Verifikasi Pengujian Otomatis (PHPUnit/Pest):

- **Standardized MRN & 16-Digit NIK:** Automated unique lifetime medical record generator RM-YYYYMM-XXXX with unique indexed national identity validation.
- **Automated Billing & Invoice Receipts:** Real-time aggregation of consultation fees and locked prescription prices with cash change computation and print-ready receipts.
- **100% Green Automated Test Suite:** 14 test cases with 77 assertions completely passed in ~1.3 seconds across full workflows, pessimistic locking logic, and multi-role auth.

5. Cybersecurity Audit Case Study: RS Online & PDO Hardening

Proyek: Sistem Informasi & Pendaftaran Pasien RS Online	**Arsitektur:** PHP • MariaDB • PDO Prepared Statements
Audit Keamanan: SAST Vulnerability Assessment • OWASP Top 10	**Regulasi Privasi:** UU Perlindungan Data Pribadi (UU PDP No. 27/2022)
Temuan Kerentanan: SQL Injection Kritis (CVSS: 9.8) Dimutasi 100%	**Pola Rekayasa:** Database Singleton & Patient Repository Pattern


TECHNICAL AUDIT & PORTFOLIO SHOWCASE

SISTEM INFORMASI & PENDAFTARAN PASIEN RUMAH SAKIT ONLINE

Laporan Analisis Arsitektur Rekayasa Perangkat Lunak, Evaluasi Skema Basis Data Relasional, Audit Celah Keamanan Siber (Vulnerability Assessment), dan Lembar Portofolio Profesional Full-Stack Web Development.

RINGKASAN INTI DOKUMEN

Dokumen ini menyajikan evaluasi mendalam terhadap aplikasi web RS Online (Tugas Besar) yang dirancang untuk memodernisasi registrasi pasien dan penyediaan informasi fasilitas kesehatan. Dilengkapi dengan evaluasi struktural, pemodelan data normalisasi, pembuktian risiko keamanan (SQL Injection, CSRF, Access Control), rekomendasi refaktorisasi PDO berstandar industri, serta portofolio kompetensi rekayasa.

PENGEMBANG / AUTHOR
Yoginara Pratema Silonu

PERAN / SPESIALISASI
Full-Stack Web Developer & Engineer

KLASIFIKASI PROYEK
Web Application (Healthcare Portal & CRUD)

STATUS & KEMAMPUAN
Functional Prototype & Production Roadmap

STACK TEKNOLOGI INTI

PHP 8.x Native | MySQL / MariaDB 10.4 | Bootstrap 5.1 & 5.3 | JavaScript ES6 | HTML5 & CSS3 | Apache HTTP Server | Git Version Control

Repository: Tugas_Besar / RS Online / Audit Keamanan & Rencana Skalabilitas — Versi 2.0 (Audit & Portfolio Edition)

RS Online | Analisis Sistem, Audit Keamanan & Portofolio Rekayasa — Halaman 1

A. Latar Belakang & Digitalisasi Pendaftaran Pasien Mandiri:

RS Online digitizes outpatient intake and polyclinic queue administration. It empowers patients to submit identities, contact records, and clinical complaints via smartphone, providing administrative staff with a centralized dashboard for real-time queue auditing.

B. Temuan Audit Kerentanan SAST (Celah Kritis SQLi CVSS 9.8):

Manual SAST audits revealed critical SQL Injection vulnerabilities in legacy procedural scripts (functions.php). Unsanitized user inputs concatenated into raw SQL queries across search and mutation functions exposed sensitive patient records to data breaches.

C. Cetak Biru Refactoring Arsitektur: Singleton PDO & Repository Pattern:

To mitigate vulnerabilities under OWASP A03:2021 and Indonesian Data Protection Law (UU PDP 27/2022), the data access layer was refactored using object-oriented architectural patterns:

- **Singleton Connection (Database.php):** Centralizes PDO database connections with UTF-8 encoding and robust PDO::ERRMODE_EXCEPTION error handling.
- **Patient Repository (PatientRepository.php):** Encapsulates data queries into isolated methods executing 100% Parameterized PDO Prepared Statements, permanently neutralizing SQL injection vectors.

D. Kepatuhan Regulasi UU PDP & Rencana Skalabilitas Enterprise:

- **Sensitive Medical Data Protection:** Safeguards national identity numbers (16-digit NIK) and patient contacts against unauthorized batch exfiltration.
- **Relational Schema Normalization:** Restructured datapasien table with unique indexed keys to eliminate patient record redundancy.
- **Scalability Roadmap:** Outlined phased enterprise hardening milestones encompassing multi-role authentication and standardized REST API endpoints.

6. High-Performance Backend Portfolio: Go (Golang 1.22) Microservices



A. StepUp Footwear E-Commerce Platform & High-Concurrency REST API:

StepUp is an e-commerce backend built to sustain high transactional concurrency. Powered by Go 1.22 and GoFiber v2 (FastHTTP core), it delivers sub-millisecond response latencies, zero-allocation routing, MongoDB Atlas cloud storage, stateless JWT authentication, and Bcrypt security.

B. ManukShop Livestock Management & OpenAPI Documentation:

Integrated livestock management and trading microservice platform in Golang, utilizing embedded NoSQL snapshot modeling, interactive OpenAPI/Swagger 3.0 documentation, and responsive Tailwind CSS interfaces.

C. Benchmarking & Metrik Kecepatan Engine Go:

Driven by FastHTTP core architecture, the GoFiber pipeline achieves 10x higher connection concurrency than standard PHP servers while maintaining stable memory consumption below 20 MB under traffic spikes.

7. Community Tech & E-Government Systems: RTConnect & UD Abang Adik



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FEATURED WEB APPLICATION PORTFOLIO + E-GOVERNMENT & COMMUNITY TECH

RTConnect – Sistem Informasi, Administrasi & Transparansi Lingkungan RT Digital

Platform digital rata kelola komunitas cerdas (Smart Community Management) berbasis web yang mengotomasi penagihan iuran warga, verifikasi pembukuan kas lingkungan, penerbitan surat pengantar resmi ber-validator publik, serta transparansi keuangan lingkungan secara real-time.

15+	3 Roles	100%	8 Feature
TABEL DATABASE RELASIONAL	RBAC + PORTAL PUBLIK	OTOMASI REKONSILIASI KAS	AUTOMATED TEST SUITES (PHPUNIT)

Tantangan & Problem Statement

- **Administrasi Konvensional Lambat:** Proses penerbitan surat pengantar warga memakan waktu berhari-hari karena bergantung pada kebesaran hati pengurus di tempat.
- **Pencatatan Kas Rentan Selisih:** Penagihan kas manual "door-to-door" sering kali terjadi pembukuannya dan memicu inkonsistensi saldo kas RT.
- **Ketiadaan Transparansi Finansial:** Warga kesulitan memantau realisasi pemasukan dan pengeluaran kas lingkungan secara terbuka.
- **Aspirasi Warga Tidak Terdata:** Keluhan fasilitas rusak sering hilang di grup obrolan tanpa ada sistem tiket dan pelacakan status tindak lanjut.

Solusi Terintegrasi RTConnect

- **E-Layanan Surat 24/7:** Warga mengajukan surat secara online, diverifikasi digital oleh Ketua RT, dan menghasilkan dokumen PDF siap unduh.
- **Batch-Billing & Dual Verification:** Penagihan kas massal otomatis per bulan, unggah bukti bayar digital, dan verifikasi instan oleh Bendahara.
- **Buku Kas Umum Otomatis:** Setiap transaksi iuran kas yang disetujui otomatis tersinkronisasi ke jurnal pembukuan umum tanpa input ganda.
- **Portal Helpdesk & Komunikasi:** Saluran aspirasi warga dengan tahapan pelacakan progres resmi serta papan informasi publik.

Arsitektur Sistem & Spesifikasi Teknologi

ENTERPRISE MVC PATTERN

Backend: PHP 8.3 & Laravel 11 | Database: MySQL 8.0 | Frontend: Blade Templating | CSS: Tailwind CSS | JS: Alpine.js
Reporting: DomPDF | Analytics: Chart.js | Testing: PHPUnit

Komponen Arsitektur	Implementasi Teknis	Manfaat bagi Sistem
Pattern MVC & Resource	Modular Controller terbagi per Role (`Admin\`, `Bendahara\`, `Warga\`, `Public\`)	Separation of Concerns (SoC), kode rapi, minim duplikasi logika bisnis.
Multi-Role RBAC Middleware	`RoleMiddleware.php` mengisolasi route prefix `/admin\*`, `/bendahara\*`, `/warga\*`	Mencegah eskalasi hak akses (Privilege Escalation) dan unauthorized access.
Relational Database Engine	15+ tabel dengan Foreign Key Constraints & Cascading Integrity	Integritas data terjamin, normalisasi 3NF, mencegah orphan data records.
Audit Activity Logging	Model `ActivityLog` mencatat mutasi data aktor beserta IP & admin	Menjamin kepatuhan audit trail (siapa, kapan, dan perubahan apa yang dibuat).

Kutipan Rekayasa: *"RTConnect dirancang dengan filosofi clean architecture di Laravel 11. Setiap mutasi finansial dan status laporan dijaga oleh validasi form ketat, transaksi atomik, dan sistem audit trail otomatis."*

Portofolio Proyek Rekayasa Perangkat Lunak – Yoginara Pratama Sitorus
Halaman 1 dari 3

A. RTConnect — Platform Administrasi & Transparansi Lingkungan RT:

RTConnect is a digital smart community governance platform automating neighborhood dues billing, real-time fiscal accounting, and official resident certificate issuance with public QR verification, structured across 15+ normalized relational tables.

B. Fitur Kunci & Dampak Sosial RTConnect:

• **Public QR Validation:** Official certificates carry unique QR codes verifiable by administrative officers without requiring account logins.

• **Real-Time Ledger Auditing:** Dues and expenditure balances are transparently logged with digital invoice receipts.

C. UD Abang Adik — Arsitektur E-Commerce Zero-Cost:

Implemented an e-commerce product catalog with zero operating costs ($0/month infrastructure), leveraging intelligent caching and fault-tolerant fallbacks achieving sub-200ms API latency.

8. Deep Learning AI Research & IoT Systems: CIT-TPRO & CO2 AirBoard

CIT-TPRO: Convolution Interactive Transformer (Riset AI)	CO2 AirBoard: Sistem Telemetri Kualitas Udara (IoT)
Laboratory: AI & Biomedical Lab ULBI (2026) **Technology:** Python, PyTorch, Linear Attention O(N) **Innovation:** Investigated hybrid Convolutional Transformer architectures with linear attention O(N) and PSD+DE feature fusion for emotion recognition on benchmark EEG datasets (DEAP, DREAMER, SEED-IV).	**Domain:** IoT & Environmental Telemetry **Hardware:** ESP32 Microcontroller, NDIR CO2 Sensor **System:** Engineered real-time environmental telemetry monitoring ambient CO2 gas concentrations with interactive web dashboards and automated threshold alert systems.

Portofolio Rekayasa Perangkat Lunak Tambahan:

Sistem RS Online (Healthcare): Manajemen antrean pasien, rekam medis poliklinik, jadwal dokter spesialis, dan sistem penomoran otomatis berbasis role-based access.	**Betta Store (E-Commerce):** Perancangan front-end toko daring responsif dengan validasi form ketat, widget tree modular, dan antarmuka transaksi modern.
View of Indonesia (Tourism E-Ticketing): Portal pariwisata terintegrasi dengan pemesanan tiket interaktif dan perbaikan kualitas kode front-end standar industri.	**Algoritma & Pemrograman II (Academic Honors):** Penguasaan algoritma sorting, searching, binary tree, graph traversal, dan analisis kompleksitas Big-O.

9. Official National Competency Certification (BNSP RI)



BADAN NASIONAL SERTIFIKASI PROFESI (BNSP REPUBLIK INDONESIA)

Qualification: First-Level Computer Network Technician (Teknisi Muda Jaringan Komputer)
Certification Body: Lembaga Sertifikasi Profesi (LSP) ULBI
Certificate No: 62090 3513 5 0000004 2026
Registration No: JIK.215.00004.2026
Issue Date: October 29, 2026 • **Validity:** 3 Years (Through October 29, 2029)

Nilai Tambah bagi Tim Rekayasa Perangkat Lunak:

Provides comprehensive knowledge of TCP/IP packet transmission, network subnetting, VLAN switch configurations, and physical cabling, ensuring software applications perform optimally across underlying network architectures.

Unit Code	SKKNI National Competency Standard Title
J.611000.001.01	Collecting Technical Requirements of Network Users
J.611000.002.01	Collecting Network Equipment Data with Appropriate Technology
J.611000.003.02	Designing Computer Network Topology
J.611000.004.01	Designing Network IP Addressing and Subnetting
J.611000.005.02	Determining Network Device Specifications
J.611000.009.02	Installing Structured Network Cabling
J.611000.012.02	Configuring Switches on Computer Networks

10. Global Credentials, Complete Skills Matrix & Career Readiness

• **Java Programming Certificate** — Oracle Academy (June 2023)	Object-Oriented Programming (OOP), Data Structures, Polymorphism & Algorithm Design
• **Java Fundamentals Certificate** — Oracle Academy (January 2023)	Java Core, Data Types, Control Structures & Computational Logic
• **MATLAB Onramp Course Completion (Score: 100%)** — MathWorks (March 2023)	Numerical Matrix Computations, Visualization & Mathematical Processing
• **Introduction to JavaScript** — Great Learning (March 2024)	DOM Scripting, Modern JavaScript ES6+ & Asynchronous Handling

Matriks Keahlian Komprehensif (Technical Competency Matrix):

Programming Languages:	PHP 8.2+ (Expert), Go / Golang 1.22, JavaScript (ES6+), Python 3.12, Bash / Shell Scripting, SQL, HTML5, CSS3
Backend Frameworks:	Laravel 12.x Engine, CodeIgniter 3 MVC, GoFiber v2 (FastHTTP), Blade Templating
Databases & Storage:	MySQL, MariaDB (3NF Normalization), Oracle Database (OCI8), MongoDB Atlas (NoSQL), Flysystem
Frontend & UI/UX:	Tailwind CSS v3, Bootstrap 5 Responsive, jQuery, DataTables, Select2, Responsive Web Design
Security & Governance:	OWASP Top 10 Mitigation (CSRF, XSS, SQLi, IDOR), ISO 31000:2018 ERM, SPI Audit Compliance
Testing & QA:	PHPUnit 11 (TDD & Automated Unit/Feature Tests), Postman API Testing, Swagger OpenAPI
Operating Systems & Network:	Linux OS (CachyOS, CLI, Shell / Bash Scripting, System Mgmt), BNSP Network Technician, TCP/IP, Switching
Tools & Methodologies:	Git, GitHub CI/CD Workflow, Composer, NPM, Vite, RESTful API Microservices, Scrum/Agile

Career Readiness & Data Integrity Statement:
All engineering specifications, efficiency benchmarks, and certifications detailed in this master dossier are fully substantiated by active production repositories and official credentials. Yoginara Pratama Sitorus, S.Tr.Kom is prepared for immediate deployment as a **Full-Stack Web Developer / Software Engineer** across onsite, hybrid, and remote engineering environments.